Mohamed Mostafa Helal

AI & Deep Learning Engineer

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Graduated 2026
Exempted from Military

Summary:

An AI & Deep Learning Engineer specialized in engineering and scaling high-impact AI applications across Large Language Models (LLMs), Agentic Systems, Computer Vision, and Real-Time Video Analytics. Experienced as Lead AI Engineer architecting and publishing Small Language Models (SLMs) for structured tool calling (BFCL v4 Leaderboard), building local hybrid RAG architectures, and deploying hardware-accelerated computer vision pipelines on multi-camera streams with real-time 30+ FPS frame ingestion. Alongside strong capabilities in full-stack architecture, API integration, and edge intelligence.

Work Experience:

Samsung Innovation Campus
(July 2024 - October 2024)

AI & Data Science Resident Intern

- Co-developed a quantitative financial forecasting framework by processing raw market data and extracting 15 specialized feature parameters (rolling standard deviations, momentum indices).
- Constructed and optimized predictive models using XGBoost, Random Forest, and SARIMA structures, systematically driving down Mean Absolute Percentage Error (MAPE) from 8.3% to 6.5%.
- Conducted extensive exploratory data analysis (EDA) across retention datasets using Pandas; built interactive analytical dashboards with Plotly and Seaborn to track operational attrition trends.

Drug Control & Addiction Prevention Fund
(2024 - 2025)

Lead Full-Stack Developer — Mutatawi' Rasmi (متطوع رسمي)

- Created the entire application ecosystem from scratch, designing responsive layouts, specialized backend logic, and structured database schema to streamline anti-drug youth initiatives in Egypt.
- Built data-driven regional risk visualizations and automated hotspot tracking maps across Egyptian governorates for focused administrative decision making.
- Implemented user engagement engines, volunteer leaderboards, mission allocation tracking, and digital milestone verification rewards.

Aitronix
(2026 - Present)

Lead AI & Computer Vision Engineer

- Lead AI Engineer on **AitroniX 1s** (2.6B Bilingual Agentic SLM on Hugging Face): fine-tuned hybrid Conv+Attention architecture; benchmarked on BFCL v4 AST (68.75% across 400 problems); optimized Q4_K_M GGUF (~1.56 GB) for mobile edge execution.
- Engineered real-time computer vision pipelines, multi-camera RTSP streaming engines, and hardware-accelerated deep learning models across surveillance and traffic domains.

Education:

Bachelor of Computer Science (2022 - 2026)

- **Misr University for Science and Technology (MUST)** — Major in Artificial Intelligence
- **Cumulative GPA:** 3.62 / 4.0 (Excellent with Honors)
- **Graduation Project:** Integrated AI Ecosystem for Arabian Horses (Grade: Excellent)
- Assisted and built multiple applied AI, Computer Vision, and full-stack software architectures.

Skills:

AI & LLM Systems:

- LLM Fine-Tuning (Unsloth, LoRA/QLoRA)
- Structured Tool Calling (<tool_call> JSON/XML, AST)
- AitroniX 1s (Hugging Face), BFCL v4 Leaderboard
- Hybrid RAG (BM25 + FAISS RRF)
- vLLM, SGLang, llama.cpp, Ollama, Transformers
- Tokenization & Prompt Engineering

Computer Vision & Multimodal:

- YOLO (v8, YOLO-World, YOLO26n)
- Vision-Language Models (Florence-2, CLIP, Moondream)
- Face Recognition (YuNet, SFace, Cosine Similarity)
- 3D CNNs, ConvNeXt, MobileNet, Transfer Learning
- OpenCV, OpenCL GPU Acceleration, RTSP Streams
- Spatial Heatmap Tracking & Speed Estimation

Languages:

- Arabic : Native / Fluent
- English : Professional / Fluent

Programming & Frameworks:

- Python (NumPy, Pandas, Scikit-learn)
- PyTorch, TensorFlow, Keras
- C#, ASP.NET Core Web API
- React.js, JavaScript (ES6+), HTML5, CSS3
- ComfyUI (Stable Diffusion), DCGANs

DataBases & DevOps:

- PostgreSQL (pgvector extension)
- SQL Server Database
- n8n Workflow Automation & Webhooks
- Docker, Git & GitHub

Social Skills:

- Team Leadership & Mentorship
- Problem Solving & Critical Thinking
- System Optimization & Latency Tuning

Courses & Certifications:

- **Samsung Innovation Campus Certified Professional in AI & Data Science (2024)** — Issued by Samsung Electronics & Ministry of Communications and Information Technology (MCIT).
- **Deep Learning & Computer Vision Specialization** — Comprehensive study in CNNs, Object Detection, and Transfer Learning.
- **Machine Learning & Advanced Statistical Modeling** — Time-series forecasting, regression, and tree-based ensembles.

Projects:

- **AitroniX 1s (AitroniXAi-1s-GGUF) — Frontier Bilingual Reasoning & Agentic Model:** Lead AI Engineer for the official GGUF release of AitroniX 1s (~2.6B params), an ultra-lightweight bilingual (Arabic/English) reasoning model on a hybrid Convolution + Attention architecture with 128K context and native `<think>` Chain-of-Thought reasoning. Achieved **68.75% strict AST match on Berkeley Function-Calling Leaderboard (BFCL v4, all 400 problems)** with 100% valid tool syntax (surpassing Qwen3.5-9B at 60.13%, Gemma-4-8B at 48.39%, and Gemma-4-2B at 35.98%). Scored **77.83 on ToolSandbox**, **85.49% on IFStruct**, **51.87% on AIME25** math, and lowest hallucination rate in class on AA Omniscience (-29.50). Optimized for private mobile edge inference (~1.56 GB) and production serving via vLLM and SGLang. (Hugging Face)
- **HyperRAG (Local Hybrid RAG Engine):** Architected a localized, high-precision hybrid retrieval system combining dense vector search (**FAISS**) with sparse lexical indexing (**BM25**), merged via **Reciprocal Rank Fusion (RRF)** to eliminate dense search entity misses. Powered locally by **llama.cpp** using Google's Gemma 4 E4B for low-latency, private responses with 800-token dynamic chunking and direct document citation tracking.
- **Aitronix Real-Time High-FPS Face Recognition Engine:** High-performance computer vision pipeline delivering single-pass grid face detection via **YuNet** (~2.5ms inference on GPU via OpenCL acceleration) and 128D deep feature embeddings via **SFace** with Cosine Similarity vector matching. Built non-blocking asynchronous multithreading with decoupled worker queues (**30+ FPS**) and a custom native LibJPEG byte decoder eliminating frame buffering delay over Wi-Fi, RTSP, and USB feeds to preserve real-time frame delivery.
- **CamScence & Spatial Traffic Analytics Deployment:** Multi-camera edge video pipeline concurrently streaming **11+ RTSP IP camera feeds** with dynamic grid switching powered by YOLO. Integrated vehicle detection, persistent multi-object tracking, automated speed estimation (km/h), and spatial heatmap visualization for roadway congestion monitoring.
- **Real-Time Surveillance Threat & Burglary Detection:** Second-by-second video surveillance pipeline combining Vision-Language Models (**Microsoft Florence-2, OpenAI CLIP**) with OpenCV for automated detection of theft, burglary, and physical violence. Utilized zero-shot visual classification and phrase grounding to score danger probabilities in real-time with automated dynamic alert banners.
- **Open-Vocabulary Real-Time Detection & Tracking (YOLO-World):** Deployed YOLO-World open-vocabulary object detector to detect and track custom arbitrary natural language prompted concepts locally on GPU at **70+ FPS** on high-speed video feeds without model retraining.
- **Dynamic WhatsApp AI Recruitment Assistant & RAG Pipeline:** Event-driven data streaming pipeline in n8n triggered by Google Sheets updates, extracting embeddings into **Supabase pgvector**. Integrated an intelligent conversational agent via Meta WhatsApp Cloud API (ngrok) executing autonomous similarity vector retrieval for applicant screening and onboarding.
- **Integrated AI Ecosystem for Arabian Horses (Graduation Project):** Architected an end-to-end multi-role web platform (React.js + ASP.NET Core REST API + SQL Server) for horse auctions, farm management (Marabet), and medical records. Integrated Python computer vision using **ConvNeXt** for horse pedigree/strain identification and price prediction.
- **NLP Architecture Benchmark on 50,000 IMDb Reviews:** Benchmarked 6 model architectures: DistilBERT (91.36%), TF-IDF + Logistic Regression (89.94%), MLP/ANN (89.66%), GRU (86.89%), BiLSTM (86.61%), SimpleRNN (82.82%). Optimized GPU training with dual Nvidia T4 GPUs and **FP16 mixed-precision**.
- **Edge Multimodal VQA & Live Video Description (Moondream):** Deployed Moondream lightweight VLM on edge hardware and Colab with live webcam integration for real-time Visual Question Answering (VQA) and continuous video description.
- **Generative AI & Synthesis Architectures:** Engineered multi-checkpoint ComfyUI node workflows with Stable Diffusion; programmed DCGAN from scratch in TensorFlow/Keras (75 epochs on Anime Faces); built 3D CNN + MobileNet-LSTM violence detection and 38-class plant disease classifier (**95.4% accuracy**).
- **Personal Resume Intelligence Assistant:** Local Retrieval-Augmented Generation (RAG) assistant querying CV records via Ollama and vector storage without reliance on external cloud APIs.